Design FMEA - VBF-T2R2-DFMEA-01

Case t2_r2 - Virtual Battery Factory - generated 2026-08-26

Design FMEA - qualitative version, S/O ratings based on simulation signals (annotation)

Failure mode	Cause	Simulation signal	Severity	Occurrence	Mitigation
Neg. electrode plating	4C areal current > anode transport	min anode potential +0.0410 V vs 0 V (final_4c_dfn.json)	High (if occurs)	Low (margin +{AP_MIN*1e3:.1f} mV)	implemented: anode rate-optimization (p 0.42, r 2.5 um) + high-sigma electrolyte + N/P > 1
Thermal runaway risk	4C heating > heat removal	T_max 360.1 K = 86.9 degC @4C/45 C, h = 10 W/m2K contract	Medium (escalation would be high)	Low (single-cycle protocol)	add cooling / limit 4C duty; acceptance already limited
Electrolyte oxidative decomposition	HOMO above cathode at top of charge	no DFT endorsement (real_compute=false, Stage-5 skipped - recorded); window 2.5-4.2 V baseline	High (if occurs)	Low-moderate (not endorsed)	keep 4.2 V cap; DFT/cV tests in physical development (flagged)
Insufficient capacity	loading/utilization below target	1C DFN 6.9089 Ah vs nominal 5.0 Ah	Low	Low (6.91 > 5.0)	none required
Excessive SEI growth	SEI kinetics too fast	99.9 nm @100 / 330.1 nm @500 at k = 2.00e-15 m/s	Medium (if exceeded)	Low at this k; k is an aggressive estimate	coating direction implemented; k must be physically realized - flagged
Coating/electrolyte realization	bridge parameters are estimates, not measured	no measured property data (annotation-level); literature-direction anchors	Medium	Medium	physical development: verify coating k and low-T sigma by measurement before production
Low-T performance loss	transport degradation at -20 C	retention 99.62 % (lumped) / 99.43 % (isothermal) vs >= 90 %	Low	Low (margin ~10 points)	keep low-T electrolyte direction (literature)

Conclusion

Highest-risk: (1) coating/electrolyte-formulation realization (estimate bridge parameters) and (2) 4C thermal (87 degC, acceptance-limited). Mitigations implemented in design; realization closed by physical development (flagged). Complete FMEA incl. process/supplier failures: N/A (beyond simulation boundary).